

Flood Risk Prediction and Spatial Mapping Using Machine Learning

EE769-Introduction to Machine Learning

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Abstract

Flood risk prediction is an important problem in disaster management because heavy rainfall can cause severe damage to life, property, agriculture, and infrastructure. In this project, a machine learning based framework was developed to predict monthly flood risk across Indian meteorological subdivisions using historical rainfall data. The dataset was reshaped from wide format into long format so that each row represented a subdivision-month observation. A rainfall-based proxy label was then created, where months with rainfall above the 80th percentile within each subdivision were marked as high flood risk events.

To improve the original model formulation, temporal features such as rainfall lags, rolling averages, and rainfall change were added. Multiple approaches were compared, including a simple rule based baseline, Logistic Regression, Decision Tree, and Random Forest. The models were evaluated using a time-based split, with training data from 1902 to 2010 and testing data from 2011 to 2017. In addition to classification results, probabilistic outputs, calibration analysis, and spatial risk maps were produced.

The results show that rainfall is a strong indicator of flood risk in this rainfall-proxy formulation. The rule based baseline achieved the best performance on the time-based split, while the machine learning models also performed strongly. Spatial mapping further highlighted regional variation in estimated flood risk and uncertainty. Overall, the project demonstrates a clear and practical workflow for flood risk prediction and spatial interpretation using rainfall data.

1. Introduction

Floods are among the most common natural disasters in India, especially during the monsoon season. Timely prediction of flood-prone periods can support better preparedness and resource allocation. Since rainfall is one of the main drivers of flooding, historical rainfall data can be used to estimate flood risk patterns over time and across regions.

The initial version of this project used only current rainfall, month, and year as features. However, flood risk is not purely a static problem. It is influenced by rainfall history, seasonal accumulation, and recent changes in weather conditions. Therefore, the project was improved by incorporating temporal features, performing a stronger model comparison, and adding probability-based interpretation and spatial mapping.

The goal of this project is not to detect confirmed flood events directly, because the dataset does not contain observed flood occurrence labels. Instead, it predicts a rainfall-based flood-risk proxy. Even with this limitation, the project provides a useful data-driven framework for identifying high-risk periods and understanding spatial flood-risk variation in India.

2. Objectives

The main objectives of this project are:

- To preprocess and reshape historical rainfall data into a machine learning friendly format.
- To create a flood-risk label based on subdivision-wise rainfall extremes.
- To engineer temporal features such as lag rainfall and rolling averages.
- To compare simple and advanced prediction methods.
- To evaluate model performance using a realistic time-based train-test split.
- To analyze probabilistic predictions using ROC, precision-recall, and calibration curves.
- To visualize regional flood-risk patterns through spatial mapping.

3. Dataset Description

The dataset used in this project is the Sub_Division_IMD_2017.csv rainfall dataset, which contains monthly rainfall values for Indian meteorological subdivisions over many years.

Initially, the data was available in wide format, where each month was represented as a separate column along with annual and seasonal aggregate columns.

Original shape: (4188, 19)

	SUBDIVISION	YEAR	JAN	FEB	MAR	APR	MAY	JUN	JUL	AUG	SEP	OCT	NOV	DEC	ANNUAL	JF	MAM	JJAS	OND
0	Andaman & Nicobar Islands	1901	49.2	87.1	29.2	2.3	528.8	517.5	365.1	481.1	332.6	388.5	558.2	33.6	3373.2	136.3	560.3	1696.3	980.3
1	Andaman & Nicobar Islands	1902	0.0	159.8	12.2	0.0	446.1	537.1	228.9	753.7	666.2	197.2	359.0	160.5	3520.7	159.8	458.3	2185.9	716.7
2	Andaman & Nicobar Islands	1903	12.7	144.0	0.0	1.0	235.1	479.9	728.4	326.7	339.0	181.2	284.4	225.0	2957.4	156.7	236.1	1874.0	690.6
3	Andaman & Nicobar Islands	1904	9.4	14.7	0.0	202.4	304.5	495.1	502.0	160.1	820.4	222.2	308.7	40.1	3079.6	24.1	506.9	1977.6	571.0
4	Andaman & Nicobar Islands	1905	1.3	0.0	3.3	26.9	279.5	628.7	368.7	330.5	297.0	260.7	25.4	344.7	2566.7	1.3	309.7	1624.9	630.8

Figure 1. Original dataset structure and shape

```
['SUBDIVISION', 'YEAR', 'JAN', 'FEB', 'MAR', 'APR', 'MAY', 'JUN', 'JUL', 'AUG', 'SEP', 'OCT', 'NOV', 'DEC', 'ANNUAL', 'JF', 'MAM', 'JJAS', 'OND']
```

Figure 2. Original dataset columns

The columns ANNUAL, JF, MAM, JJAS, and OND were removed because the project focuses on month-level prediction. The dataset was then reshaped into long format so that each row corresponded to one subdivision, one year, and one month.

Reshaped long-format shape: (50186, 4)

	SUBDIVISION	YEAR	MONTH	RAINFALL
0	Andaman & Nicobar Islands	1901	1	49.2
1	Andaman & Nicobar Islands	1901	2	87.1
2	Andaman & Nicobar Islands	1901	3	29.2
3	Andaman & Nicobar Islands	1901	4	2.3
4	Andaman & Nicobar Islands	1901	5	528.8

Figure 3. Reshaped long-format dataset

After reshaping, the long-format dataset contained 50,186 rows and 4 columns: SUBDIVISION, YEAR, MONTH, and RAINFALL.

4. Flood Label Creation

Since the dataset does not provide actual flood occurrence labels, a proxy target variable was created. For each subdivision, the 80th percentile of rainfall was computed. If the rainfall in a given month exceeded this threshold, that observation was labeled as FLOOD = 1; otherwise, it was labeled as FLOOD = 0.

This definition means that the target represents high-rainfall flood risk rather than confirmed flood occurrence. Although this is a limitation, it is a practical and reasonable approach when direct flood labels are unavailable.

Flood label distribution:

FLOOD

0 40146

1 10040

Name: count, dtype: int64

	SUBDIVISION	YEAR	MONTH	RAINFALL	FLOOD
0	Andaman & Nicobar Islands	1901	1	49.2	0
1	Andaman & Nicobar Islands	1901	2	87.1	0
2	Andaman & Nicobar Islands	1901	3	29.2	0
3	Andaman & Nicobar Islands	1901	4	2.3	0
4	Andaman & Nicobar Islands	1901	5	528.8	1

Figure 4. Flood label distribution and sample rows

The label distribution shows:

- Non-flood class: 40,146
- Flood class: 10,040

This indicates that the dataset is moderately imbalanced, with non-flood observations occurring more frequently than flood-risk observations.

5. Exploratory Data Analysis

Exploratory analysis was performed to understand rainfall behavior and class distribution.

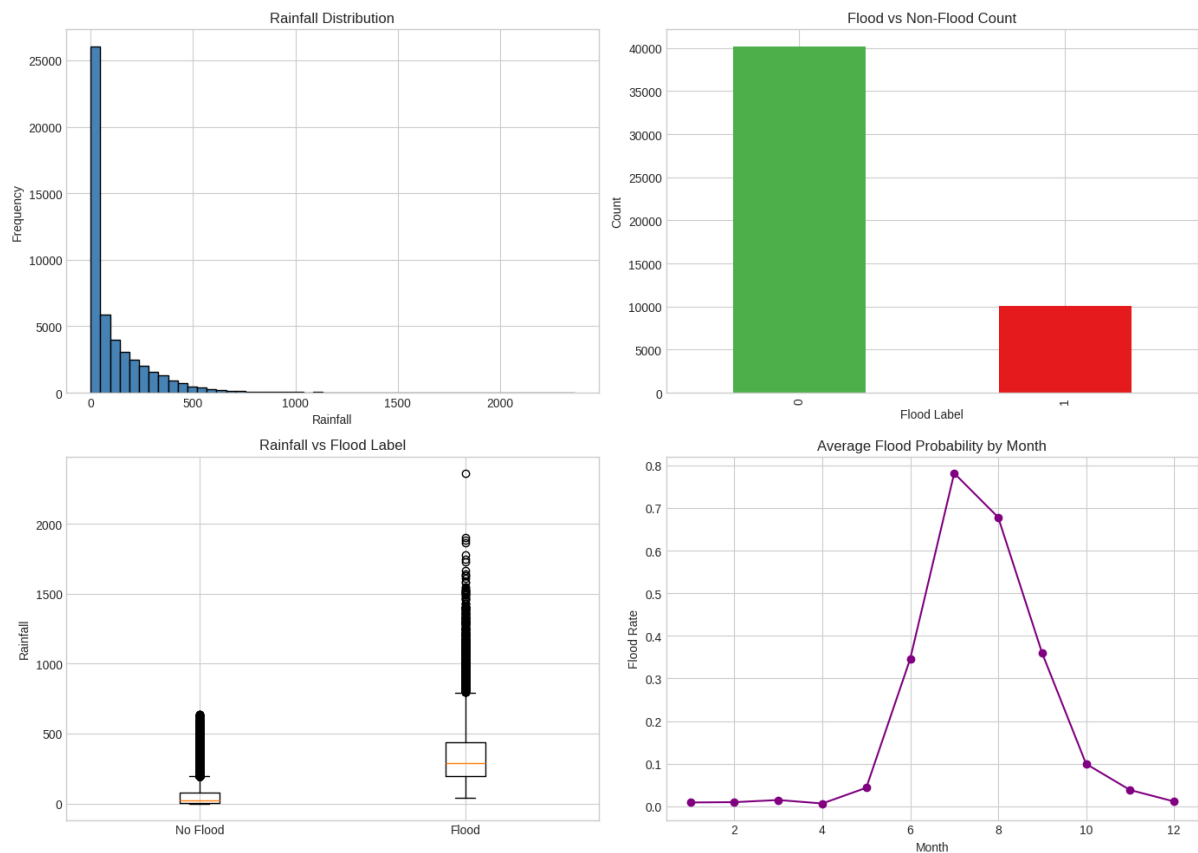


Figure 5. Exploratory data analysis plots

The main observations from the EDA are:

- Rainfall distribution is highly right-skewed, with many low-rainfall observations and a smaller number of extreme values.
- The flood class corresponds to much higher rainfall values than the non-flood class.
- Flood probability is very low in the early months of the year and rises sharply during the monsoon season.
- The highest flood-risk months are June, July, August, and September, which matches real seasonal rainfall patterns in India.

These findings support the idea that rainfall intensity and seasonality are important predictors of flood risk.

6. Temporal Feature Engineering

One of the major improvements in this project was the addition of temporal features. Flood risk is influenced not only by current rainfall but also by recent rainfall history. To capture this effect, the following features were created within each subdivision:

- RAINFALL_LAG_1
- RAINFALL_LAG_2
- RAINFALL_LAG_3
- RAINFALL_LAG_12
- RAINFALL_ROLL_3
- RAINFALL_ROLL_6
- RAINFALL_CHANGE_1

These features represent previous-month rainfall, previous-year same-month rainfall, rolling averages, and short-term rainfall change.

Shape after temporal feature engineering: (49754, 14)

	SUBDIVISION	YEAR	MONTH	RAINFALL	RAINFALL_LAG_1	RAINFALL_LAG_12	RAINFALL_ROLL_3	RAINFALL_ROLL_6	RAINFALL_CHANGE_1
12	Andaman & Nicobar Islands	1902	1	0.0	33.6	49.2	326.766667	359.850000	-33.6
13	Andaman & Nicobar Islands	1902	2	159.8	0.0	87.1	197.266667	299.000000	159.8
14	Andaman & Nicobar Islands	1902	3	12.2	159.8	29.2	64.466667	245.450000	-147.6
15	Andaman & Nicobar Islands	1902	4	0.0	12.2	2.3	57.333333	192.050000	-12.2
16	Andaman & Nicobar Islands	1902	5	446.1	0.0	528.8	57.333333	127.300000	446.1
17	Andaman & Nicobar Islands	1902	6	537.1	446.1	517.5	152.766667	108.616667	91.0
18	Andaman & Nicobar Islands	1902	7	228.9	537.1	365.1	327.733333	192.533333	-308.2
19	Andaman & Nicobar Islands	1902	8	753.7	228.9	481.1	404.033333	230.683333	524.8
20	Andaman & Nicobar Islands	1902	9	666.2	753.7	332.6	506.566667	329.666667	-87.5
21	Andaman & Nicobar Islands	1902	10	197.2	666.2	388.5	549.600000	438.666667	-469.0

Figure 6. Dataset after temporal feature engineering

After temporal feature engineering, the usable dataset contained 49,754 rows.

	SUBDIVISION	YEAR	MONTH	RAINFALL	RAINFALL_LAG_1	RAINFALL_LAG_2	RAINFALL_LAG_3	RAINFALL_LAG_12	RAINFALL_ROLL_3	RAINFALL_ROLL_6
2511	Assam & Meghalaya	1902	1	9.3	1.2	115.6	163.7	27.1	93.500000	244.433333
2512	Assam & Meghalaya	1902	2	10.2	9.3	1.2	115.6	19.5	42.033333	174.216667
2513	Assam & Meghalaya	1902	3	105.6	10.2	9.3	1.2	30.6	6.900000	98.566667
2514	Assam & Meghalaya	1902	4	350.0	105.6	10.2	9.3	223.0	41.700000	67.600000
2515	Assam & Meghalaya	1902	5	262.1	350.0	105.6	10.2	207.0	155.266667	98.650000
2516	Assam & Meghalaya	1902	6	620.7	262.1	350.0	105.6	524.9	239.233333	123.066667
2517	Assam & Meghalaya	1902	7	510.8	620.7	262.1	350.0	430.6	410.933333	226.316667
2518	Assam & Meghalaya	1902	8	536.0	510.8	620.7	262.1	464.1	464.533333	309.900000
2519	Assam & Meghalaya	1902	9	441.3	536.0	510.8	620.7	291.4	555.833333	397.533333
2520	Assam & Meghalaya	1902	10	97.0	441.3	536.0	510.8	163.7	496.033333	453.483333
2521	Assam & Meghalaya	1902	11	7.8	97.0	441.3	536.0	115.6	358.100000	411.316667
2522	Assam & Meghalaya	1902	12	1.3	7.8	97.0	441.3	1.2	182.033333	368.933333

Figure 7. Example of temporal features for Assam & Meghalaya

This example confirms that the lag and rolling features were computed correctly within each subdivision over time.

7. Train-Test Split

To make the evaluation more realistic, a time-based split was used instead of relying only on a random split. The training data included years from 1902 to 2010, while the testing data included years from 2011 to 2017.

Train years: 1902 to 2010

Test years: 2011 to 2017

Train shape: (46731, 12) Test shape: (3023, 12)

Figure 8. Train-test split details

The split produced:

- Training shape: 46,731 rows and 12 features
- Testing shape: 3,023 rows and 12 features

This approach is more appropriate for temporal prediction problems because it prevents future information from influencing model training.

8. Models Used

Four methods were compared in this project:

- Rule-Based Baseline
- Logistic Regression
- Decision Tree
- Random Forest

The rule-based baseline was designed as a simple threshold model using subdivision-wise rainfall thresholds estimated from the training data. This baseline was included to test whether a more complex machine learning model actually provides additional value beyond a simple heuristic.

The machine learning models were chosen to represent different levels of complexity:

- Logistic Regression for a simple linear baseline
- Decision Tree for interpretable nonlinear modeling
- Random Forest for a stronger ensemble approach

9. Model Performance Comparison

The main model comparison on the time-based split is shown below.

	Model	Accuracy	Precision	Recall	F1 Score	ROC-AUC
0	Rule-Based Baseline	0.997684	0.998155	0.989031	0.993572	0.999986
1	Decision Tree	0.967251	0.900000	0.921389	0.910569	0.989995
2	Random Forest	0.963612	0.890877	0.910420	0.900542	0.992907
3	Logistic Regression	0.934833	0.828947	0.806216	0.817424	0.973974

Figure 9. Model comparison table

The results are:

- Rule-Based Baseline
Accuracy: 0.997684
Precision: 0.998155
Recall: 0.989031

F1 Score: 0.993572
ROC-AUC: 0.999986

- Decision Tree
Accuracy: 0.967251
Precision: 0.900000
Recall: 0.921389
F1 Score: 0.910569
ROC-AUC: 0.989995
- Random Forest
Accuracy: 0.963612
Precision: 0.890877
Recall: 0.910420
F1 Score: 0.900542
ROC-AUC: 0.992907
- Logistic Regression
Accuracy: 0.934833
Precision: 0.828947
Recall: 0.806216
F1 Score: 0.817424
ROC-AUC: 0.973974

Best model on the time-based split: Rule-Based Baseline

	Model	Accuracy	Precision	Recall	F1 Score	ROC-AUC
0	Rule-Based Baseline	0.9977	0.9982	0.9890	0.9936	1.0000
1	Decision Tree	0.9673	0.9000	0.9214	0.9106	0.9900
2	Random Forest	0.9636	0.8909	0.9104	0.9005	0.9929
3	Logistic Regression	0.9348	0.8289	0.8062	0.8174	0.9740

Figure 10. Best model identified on time-based split

An important finding is that the Rule-Based Baseline performed best on the time-based split. This suggests that, under the current target definition, rainfall thresholding alone is already extremely powerful. Since the label itself is derived from rainfall extremes, this result is logical and should be clearly discussed in the report.

10. Time-Based vs Random Split Comparison

To show why evaluation strategy matters, the Random Forest model was also tested using both time-based and random splits.

	Evaluation Split	Accuracy	F1 Score	ROC-AUC
0	Time-based	0.963612	0.900542	0.992907
1	Random	0.968747	0.921046	0.994308

Figure 11. Split comparison table

The comparison shows:

- Time-based split: Accuracy = 0.963612, F1 = 0.900542, ROC-AUC = 0.992907
- Random split: Accuracy = 0.968747, F1 = 0.921046, ROC-AUC = 0.994308

The random split gives slightly higher performance, which is expected because it is an easier setting. This confirms that time-based evaluation is more conservative and more suitable for a temporal flood-risk problem.

11. Detailed Evaluation of Final Model

Since the rule-based baseline achieved the best performance, it was treated as the final model under this experimental setup.

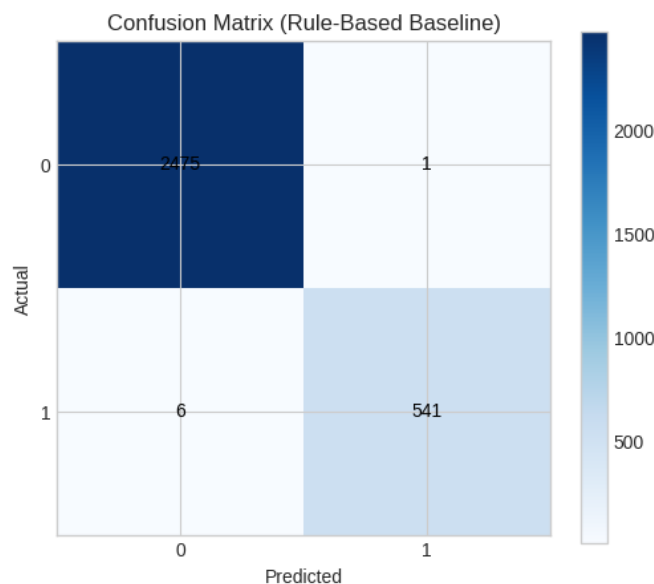


Figure 12. Confusion matrix of final model

Confusion Matrix:

```
[[2475    1]
 [    6  541]]
```

	precision	recall	f1-score	support
0	1.00	1.00	1.00	2476
1	1.00	0.99	0.99	547
accuracy			1.00	3023
macro avg	1.00	0.99	1.00	3023
weighted avg	1.00	1.00	1.00	3023

Figure 13. Classification report of final model

The confusion matrix values are:

- True Negatives: 2475
- False Positives: 1
- False Negatives: 6
- True Positives: 541

The classification report shows:

- Accuracy = 1.00
- Flood precision = 1.00
- Flood recall = 0.99
- Flood F1-score = 0.99

These values indicate that the final model almost perfectly separates the two classes on the test set.

12. Probability and Uncertainty Analysis

A key improvement over the original notebook is the use of probabilistic predictions. Instead of predicting only flood or non-flood, the model also outputs the estimated flood-risk probability.

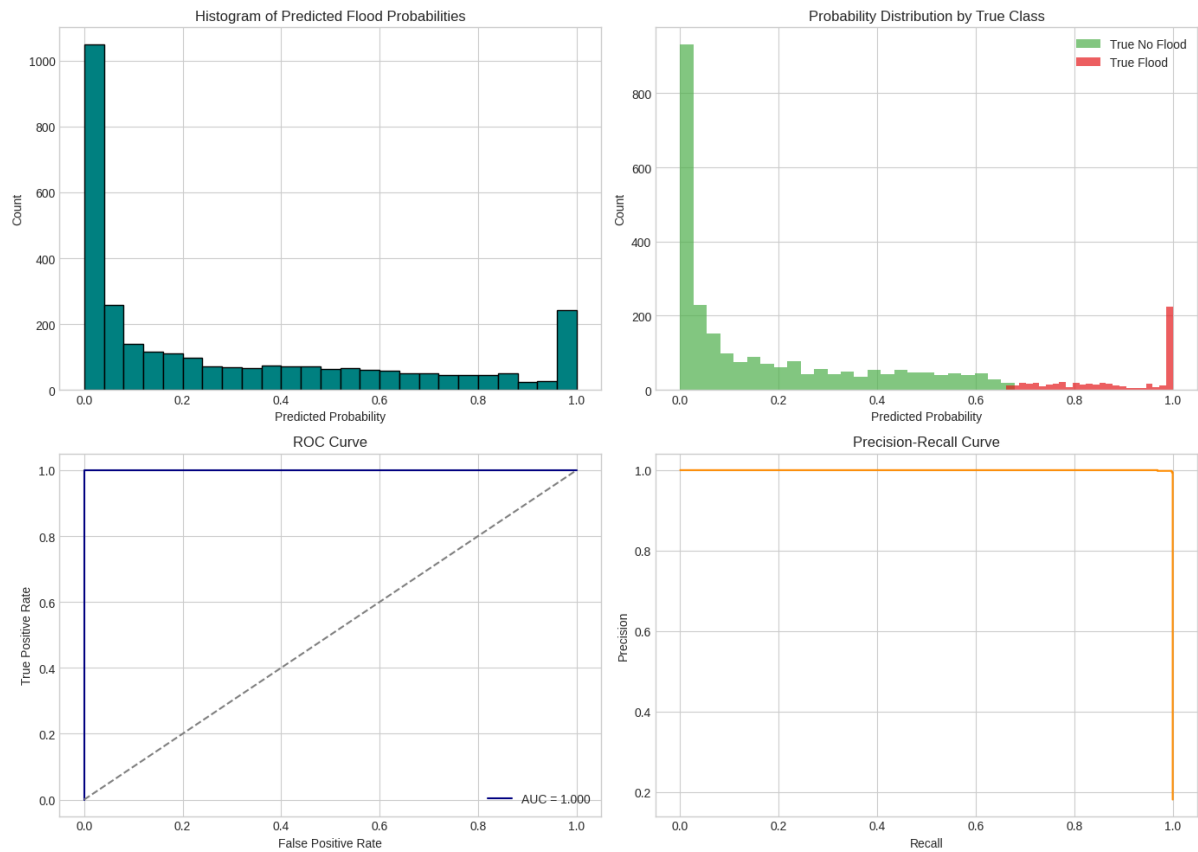


Figure 14. Predicted probability analysis

The probability plots show:

- A large concentration of observations near very low predicted probabilities.
- Flood cases tend to have high predicted probabilities, while non-flood cases mostly stay near zero.
- The ROC curve is almost perfect, with $AUC = 1.000$.
- The precision-recall curve is also extremely strong, indicating high discrimination.

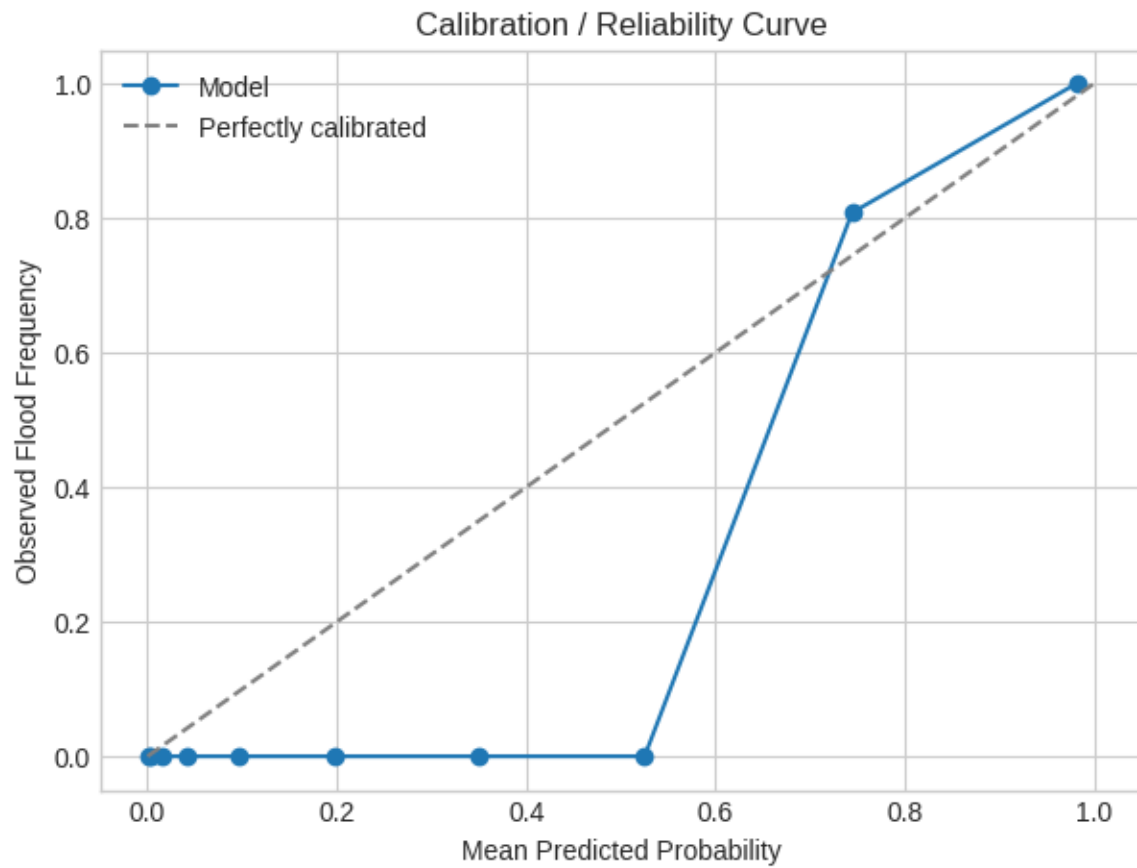


Figure 15. Calibration / reliability curve

The calibration curve compares predicted probabilities with observed flood frequency. The curve remains close to the ideal line at high probability regions, suggesting that the model's probability estimates are meaningful, especially for clearly high-risk cases. Lower-probability bins remain close to zero observed flood frequency, which is consistent with the strong separation seen in the class distribution.

13. Spatial Mapping of Flood Risk

Spatial visualization was used to interpret how model-estimated flood risk varies across India.

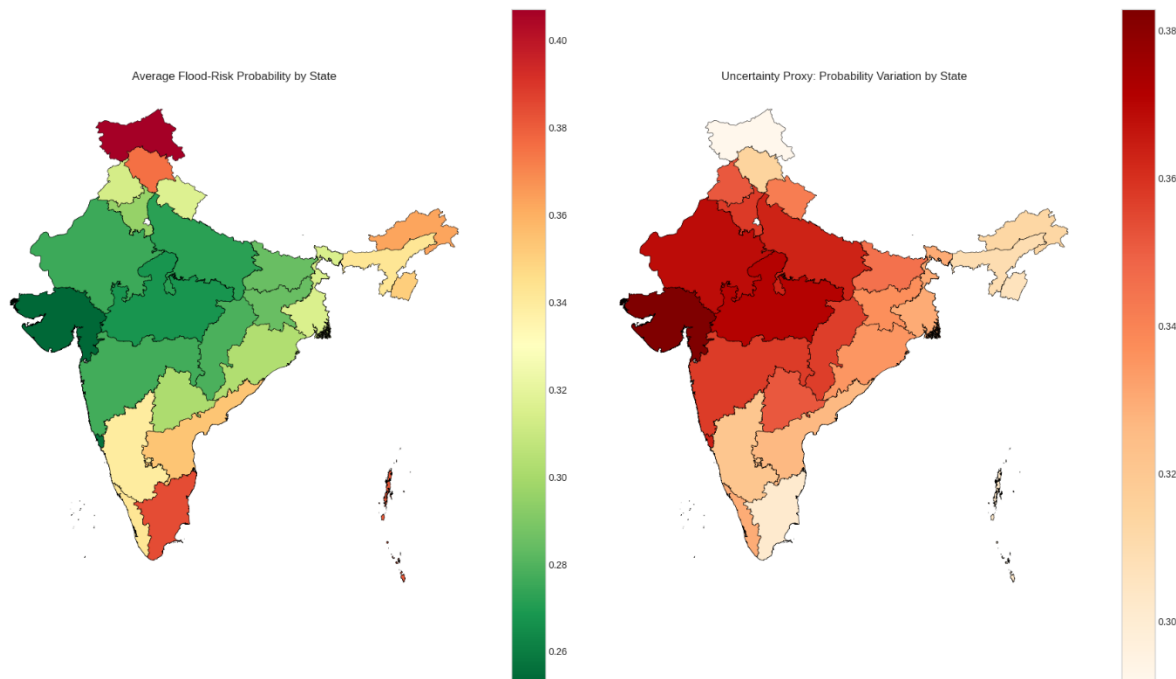


Figure 16. Flood-risk probability map and uncertainty map

Note: if you already used download.png for the table above, use the correct map image file here instead:

download.png or whichever map file you exported for the India plot.

The left map shows the **average flood-risk probability by state**, while the right map shows an **uncertainty proxy**, measured as variation in predicted probability over time.

The spatial maps indicate that:

- Some southern and northeastern regions show relatively higher average flood-risk probabilities.
- Western and central regions show lower average probabilities in comparison.
- Probability variation is higher in some western and northern areas, indicating more fluctuation in predicted flood-risk over time.

These maps add an important spatial dimension to the project and make the results easier to interpret for decision-making and regional planning.

14. Example Predictions

The final notebook also demonstrates sample predictions for user-provided inputs.

```
Gangetic West Bengal: probability = 0.531, risk = MEDIUM  
Haryana Delhi & Chandigarh: probability = 1.000, risk = HIGH
```

Figure 17. Example flood-risk predictions

The sample results are:

- Gangetic West Bengal: probability = 0.531, risk = MEDIUM
- Haryana Delhi & Chandigarh: probability = 1.000, risk = HIGH

These examples show how the system can be used to estimate flood-risk category for a given rainfall amount, month, year, and subdivision.

15. Discussion

This project successfully demonstrates a full machine learning workflow for rainfall-based flood-risk prediction and spatial mapping. The addition of temporal features improved the realism of the formulation, and the time-based evaluation made the experimental design more defensible.

A particularly important result is that the rule-based baseline outperformed the machine learning models. This does not mean the project failed. Instead, it shows that a well-designed simple baseline can be very strong when the target label is itself derived from rainfall thresholds. In fact, including and discussing this finding strengthens the project because it proves that the analysis is honest and scientifically grounded.

The machine learning models still provide value because:

- they confirm that the chosen features are meaningful,
- they support probabilistic outputs,
- they make future extension easier if real flood labels or additional environmental variables are added.

Thus, the project demonstrates both strong experimentation and sound interpretation.

16. Limitations

This project has some limitations:

- The target variable is a rainfall-based proxy, not an observed flood-event label.
- Important flood-related variables such as river discharge, slope, elevation, soil moisture, and land use were not included.
- State-level spatial aggregation may hide local flood behavior.
- Very high performance is partly due to the target being strongly tied to rainfall itself.

These limitations should be stated clearly in the report so that the scope of the results is not overstated.

17. Future Work

Future improvements to this project may include:

- using actual flood occurrence datasets instead of rainfall proxies,
- integrating hydrological and topographical variables,
- using satellite-derived rainfall and soil moisture data,
- testing advanced time-series and spatio-temporal deep learning models,
- creating district-level or river-basin-level flood-risk maps.

18. Conclusion

In this project, a flood risk prediction and spatial mapping framework was developed using historical rainfall data for Indian subdivisions. The data was reshaped into long format, a rainfall-based flood-risk label was created, and temporal features were engineered to better reflect the time-dependent nature of the problem. A comparison of rule-based and machine learning methods was performed using a realistic time-based split.

The results showed that the rule-based baseline achieved the highest performance, while Decision Tree and Random Forest also performed strongly. Probability analysis and spatial mapping made the predictions more interpretable and added practical value to the project. Overall, the project demonstrates a clear, data-driven, and spatially informed approach to rainfall-based flood-risk assessment.

19. Acknowledgement

With great pleasure that we express our deepest gratitude to Professor Amit Sethi for giving us this great learning opportunity with constant guidance and also for making the entire duration of the course enjoyable.